TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	58.3 ± 3.86
Correlation Stability	81.7 ± 3.09
Distribution Consistency	69.0 ± 4.90
Entropy Preservation	77.7 ± 2.36
Mean Z Score Drift	0.2 ± 0.07
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	26.3 ± 2.49
Data Density	100.0 ± 0.00
Coverage Score	66.3 ± 4.78
Feature Accuracy	87.0 ± 4.32
Novelty Score	0.6 ± 0.06

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1-nano
Folds (*13)	3
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	162.254	+8.026	5.2%	OK
MDVP:Fhi	197.105	189.300	-7.805	4.0%	OK
MDVP:Flo	116.325	115.338	-0.987	0.8%	OK
MDVP:Jitter	0.006	0.008	+0.002	26.4%	Drift
MDVP:Jitter.1	0.006	0.008	+0.002	26.7%	Drift
MDVP:RAP	0.003	0.004	+0.001	31.1%	Drift
MDVP:PPQ	0.003	0.004	+0.001	29.7%	Drift
Jitter:DDP	0.010	0.013	+0.003	29.2%	Drift
MDVP:Shimmer	0.030	0.040	+0.010	34.8%	Drift
MDVP:Shimmer.1	0.030	0.040	+0.011	35.5%	Drift
Shimmer:APQ3	0.016	0.019	+0.004	23.4%	Warn
Shimmer:APQ5	0.018	0.025	+0.007	41.1%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	58.3 ± 3.86
Correlation Stability (*2)	81.7 ± 3.09
Distribution Consistency (*3)	69.0 ± 4.90
Entropy Preservation (*4)	77.7 ± 2.36
Mean Z Score Drift (*5)	0.2 ± 0.07
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	26.3 ± 2.49

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	66.3 ± 4.78
Feature Accuracy (*10)	87.0 ± 4.32
Novelty Score (*11)	0.6 ± 0.06

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.8290598290598291	0.9743589743589743	0	
F1	0.980643539467069	0.8827439807831965	0.980643539467069	0	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8717948717948718	0.7264957264957265	0.8803418803418803	0.008547008547008517	
F1	0.9037242015450248	0.8220469520639108	0.9054232804232804	0.0016990788878255599	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.94017094017094	0.8119658119658121	0.94017094017094	0	
F1	0.9563061353573689	0.8712218130128577	0.9578948959845186	0.0015887606271498915	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	0.7948717948717948	0.94017094017094	-0.01709401709401707	
F1	0.968042328042328	0.8627384960718295	0.9544997486173957	-0.013542579424932315	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.7521367521367521	0.7948717948717948	0.01709401709401707	
F1	0.8614267676767677	0.8482424720738222	0.8710053117380703	0.009578544061302652	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.7264957264957266	0.8803418803418803	3.700743415417188e-17	
F1	0.91232423490488	0.8167055167055167	0.9174545580205958	0.005130323115715707	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.7435897435897436	0.8376068376068376	-0.008547008547008553	
F1	0.8918089248277927	0.8376157099831695	0.8891342575553102	-0.002674667272482593	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.7948717948717948	0.7094017094017094	-0.008547008547008553	
F1	0.749906191369606	0.8457978802806388	0.7437118437118437	-0.006194347657762315	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7692307692307693	0.7350427350427351	0.8547008547008547	0.08547008547008539	
F1	0.8543140377516881	0.8324910885063468	0.899204021784667	0.04488998403297872	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.8803418803418803	0.9230769230769231	-0.017094017094017106	
F1	0.9570703384745131	0.9148752482085816	0.9429141229141229	-0.014156215560390187	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	228.832	234.619	223.634	0.00296	0.00296	0.00175	0.00155	0.00526	0.01644	0.01644
2	116.388	129.038	108.97	0.00346	0.00346	0.00169	0.00213	0.00507	0.01725	0.01725
3	150.258	154.609	75.349	0.00248	0.00248	0.00105	0.00136	0.00315	0.01131	0.01131
4	197.569	217.627	90.794	0.00803	0.00803	0.0049	0.00448	0.0147	0.02177	0.02177
5	142.729	162.408	65.476	0.00831	0.00831	0.00469	0.00419	0.01407	0.03485	0.03485
6	125.791	140.557	96.206	0.01378	0.01378	0.00826	0.00655	0.02478	0.04689	0.04689
7	195.4	220.1	80.5	0.0032	0.0032	0.0018	0.0016	0.0054	0.0172	0.0172
8	162.8	180.3	75.2	0.0029	0.0029	0.0016	0.0014	0.0048	0.0149	0.0149
9	210.5	240.7	85.3	0.0035	0.0035	0.002	0.0018	0.006	0.0185	0.0185
10	185.2	210.4	78.9	0.0031	0.0031	0.0017	0.0015	0.0052	0.0163	0.0163
11	170.7	195.2	72.4	0.0028	0.0028	0.0015	0.0013	0.0045	0.0142	0.0142
12	200.1	225.9	82.1	0.0034	0.0034	0.0019	0.0017	0.0058	0.0178	0.0178
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	150.44	163.441	144.736	0.00396	0.00396	0.00206	0.00233	0.00619	0.02551	0.02551
2	178.222	202.45	141.047	0.00321	0.00321	0.00163	0.00194	0.00488	0.03759	0.03759
3	192.818	224.429	168.793	0.03107	0.03107	0.018	0.01958	0.05401	0.11908	0.11908
4	156.239	195.107	79.82	0.00694	0.00694	0.00403	0.00395	0.01209	0.02343	0.02343
5	117.274	129.916	110.402	0.00752	0.00752	0.00299	0.00469	0.00898	0.02293	0.02293
6	115.322	135.738	107.802	0.00619	0.00619	0.00352	0.00329	0.01057	0.03273	0.03273
7	157.89	198.45	135.67	0.0038	0.0038	0.0021	0.0024	0.0059	0.028	0.028
8	164.32	210.75	142.89	0.0042	0.0042	0.0023	0.0026	0.0064	0.031	0.031
9	150.45	192.33	125.44	0.0035	0.0035	0.0019	0.0022	0.0051	0.026	0.026
10	170.12	220.45	150.78	0.0045	0.0045	0.0025	0.0028	0.0071	0.035	0.035
11	155.67	200.12	130.56	0.0039	0.0039	0.0021	0.0024	0.006	0.029	0.029
12	162.45	205.67	138.89	0.0041	0.0041	0.0022	0.0025	0.0062	0.03	0.03
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

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B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	153.848	165.738	65.782	0.0084	0.0084	0.00428	0.0045	0.01285	0.0381	0.0381
2	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995
3	178.285	442.824	82.063	0.00462	0.00462	0.00157	0.00194	0.00472	0.01279	0.01279
4	197.569	217.627	90.794	0.00803	0.00803	0.0049	0.00448	0.0147	0.02177	0.02177
5	176.281	227.381	125.61	0.0052	0.0052	0.00287	0.00312	0.00862	0.06511	0.06511
6	157.821	172.975	68.401	0.00358	0.00358	0.00196	0.00196	0.00587	0.03716	0.03716
7	160.45	180.23	70.12	0.0068	0.0068	0.0035	0.0038	0.0102	0.045	0.045
8	155.78	170.45	75.34	0.0059	0.0059	0.0031	0.0033	0.0098	0.038	0.038
9	165.12	190.67	65.89	0.0072	0.0072	0.0038	0.0041	0.0115	0.05	0.05
10	158.34	175.89	80.45	0.0061	0.0061	0.0032	0.0034	0.0095	0.037	0.037
11	162.89	185.23	68.76	0.0065	0.0065	0.0034	0.0037	0.01	0.043	0.043
12	159.45	178.56	72.34	0.0058	0.0058	0.003	0.0032	0.0092	0.039	0.039
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645